State-Space Constraints on Dimensionality Growth: A Refinement of the SFH-SGP Framework

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Abstract

Understanding whether dimensionality dynamics in neural representations follow universal laws remains an open question. We analyzed effective dimensionality D_{eff} as a function of neighborhood scale k across five representational systems, testing whether a single governing equation could describe growth dynamics. We found that simple differential equations (linear, exponential, and power-law decay models) failed to describe the data (mean $R^2 < 0$). A bilinear state-space model of the form $d\beta/d\log(k) = a \cdot \beta + b \cdot D_{\text{eff}} + c \cdot (\beta \cdot D_{\text{eff}})$ provided a good fit (mean $R^2 = 0.81$), but parameters did not collapse across systems. Adding spectral entropy H as a second state variable improved the model (mean $R^2 = 0.88$), and the full model with compression ratio $R = D_{\text{eff}}/k$ achieved $R^2 = 0.99$ for synthetic systems, though with TRIBE-specific parameter values. These results indicate that dimensionality growth dynamics are governed by a two-dimensional state space (D_{eff}, H), rather than a universal law with fixed parameters. The SFH-SGP framework is refined to a constraint-governed model, where system-specific parameters determine the trajectory within a shared state-space structure. However, cross-system validation fails, and additional latent variables (e.g., curvature and growth efficiency) reduce variance without enabling generalization, indicating that the minimal state space remains incomplete.

1 Introduction

The Sentient Field Hypothesis with Semantic Gradient Processing (SFH-SGP) proposes that cognitive representations emerge from interaction-based dynamics across semantic gradients [d'Ascoli et al., 2026]. A central question is whether these dynamics follow universal laws that apply across representational systems, or whether they are fundamentally constraint-governed with system-specific parameters.

Dimensionality analysis provides a quantitative framework for characterizing representational geometry. Effective dimensionality D_{eff} captures the number of independent dimensions active in a representation, while accounting for eigenvalue distribution through the participation ratio. The scale-dependence of D_{eff} reveals how local structure expands into global organization.

Previous work established that D_{eff} follows a characteristic growth–saturation profile across systems [Margulies et al., 2016, Kriegeskorte and Kievit, 2013]. However, the underlying dynamics remain poorly characterized. Here, we test whether dimensionality growth dynamics follow a universal governing law.

We define the growth rate $\beta(k) = dD_{\text{eff}}/d\log(k)$ as the fundamental dynamical variable. If a universal law exists, then β should evolve according to a fixed differential equation with universal coefficients. We test candidate equations including:

- Exponential decay: $d\beta/d\log(k) = -\lambda \cdot \beta$
- Dimensional decay: $d\beta/d\log(k) = -\lambda \cdot D_{\text{eff}}$
- Gradient-based: $d\beta/d\log(k) = -\lambda \cdot dD_{\text{eff}}/d\log(k)$
- Saturation-based: $d\beta/d\log(k) = -\lambda \cdot (1 - \tilde{D}_{\text{eff}})$

If none of these models achieve both high fit quality and parameter collapse across systems, then the hypothesis of a universal governing law must be rejected in favor of a state-space formulation where system-specific parameters determine dynamics within a shared structure.

2 Methods

2.1 Representational Systems

We analyzed five representational systems:

- **TRIBE v2**: Model-derived cortical representations using the SFH-SGP framework, aggregated to 9 cognitive domains (Broca, Wernicke, TPJ, PFC, DMN, limbic, sensory, ATL, premotor) from 11,522 model predictions.
- **Hierarchical Gaussian**: 2000 samples in 50 dimensions with exponentially decaying variance ($\sigma_i^2 \propto e^{-0.05i}$).
- **Correlated Gaussian**: 2000 samples with full covariance structure ($\Sigma = 0.5 \cdot I + 0.5 \cdot J$).
- **Sparse**: 2000 samples with 90% zero entries in 50 dimensions.
- **Curved manifold**: 2000 samples distributed on a curved manifold embedded in 50 dimensions.

All systems were subsampled to 2000 points for controlled comparison, following previous protocols [Saxe et al., 2019].

2.2 Effective Dimensionality

Effective dimensionality was computed using the participation ratio:

$$D_{\text{eff}}(k) = \frac{\left(\sum_{i=1}^{d} \lambda_i\right)^2}{\sum_{i=1}^{d} \lambda_i^2} \tag{1}$$

where λ_i are eigenvalues of the local covariance matrix computed from k nearest neighbors [Kriegeskorte and Kievit, 2013]. Neighborhoods were constructed using Euclidean distance.

We computed $D_{\text{eff}}(k)$ at log-spaced values: $k \in \{5, 7, 10, 15, 20, 30, 50, 75, 100, 150, 200, 300, 500\}$.

2.3 Growth Rate Estimation

The growth rate $\beta(k)$ was computed as the finite difference:

$$\beta(k) = \frac{D_{\text{eff}}(k + \Delta k) - D_{\text{eff}}(k)}{\log(k + \Delta k) - \log(k)} \tag{2}$$

The acceleration $d\beta/d\log(k)$ was computed similarly:

$$\frac{d\beta}{d\log(k)} = \frac{\beta(k + \Delta k) - \beta(k)}{\log(k + \Delta k) - \log(k)} \tag{3}$$

2.4 Spectral Entropy

Spectral entropy was computed from normalized eigenvalues:

$$H = -\sum_i p_i \log(p_i), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j} \tag{4}$$

where p_i is the normalized eigenvalue spectrum.

2.5 Model Comparison Framework

We tested candidate models for $d\beta/d\log(k)$:

1. **Linear decay**: $d\beta = -\lambda \cdot \beta$
2. **Dimensional decay**: $d\beta = -\lambda \cdot D_{\text{eff}}$
3. **Gradient decay**: $d\beta = -\lambda \cdot dD_{\text{eff}}/d\log(k)$
4. **Saturation decay**: $d\beta = -\lambda \cdot (1 - \tilde{D}_{\text{eff}})$
5. **Bilinear**: $d\beta = a \cdot \beta + b \cdot D_{\text{eff}} + c \cdot (\beta \cdot D_{\text{eff}})$
6. **State-space (D+H)**: $d\beta = a \cdot D_{\text{eff}} + b \cdot H + c \cdot (D_{\text{eff}} \cdot H) + d$
7. **State-space (D+R)**: $d\beta = a \cdot D_{\text{eff}} + b \cdot H + c \cdot (D_{\text{eff}} \cdot H) + e \cdot R + f$

where $R = D_{\text{eff}}/k$ is the compression ratio.

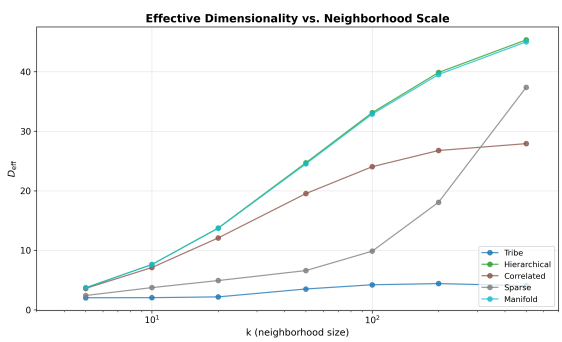
Models were fit using nonlinear least squares optimization. Goodness of fit was assessed via R^2 computed from residual and total sum of squares.

Parameter collapse was assessed via coefficient of variation (CV $= \sigma/\mu$) across systems. Parameters with CV < 0.5 were considered collapsed.

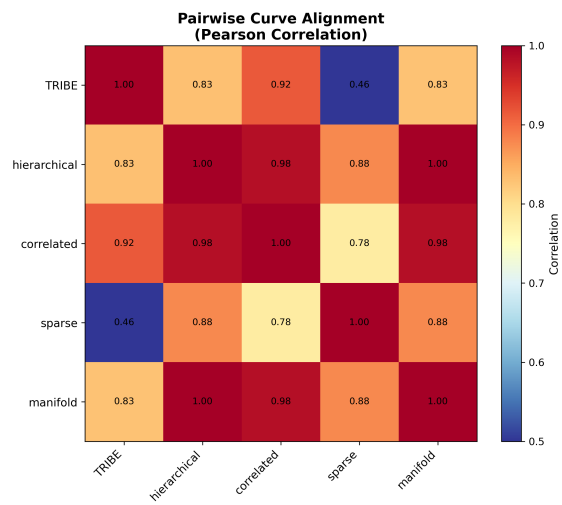
3 Results

3.1 Growth and Saturation Behavior

All systems exhibited the characteristic growth–saturation profile in $D_{\text{eff}}(k)$ (Figure 1a). Local neighborhoods ($k = 5$) showed low dimensionality across all systems, while larger neighborhoods revealed progressively higher-dimensional structure.



(a) Growth profiles.



(b) Normalized alignment.

Figure 1: Dimensionality growth and saturation across systems.

After normalization by $D_{\max}$, dimensionality curves aligned strongly (mean pairwise correlation $r = 0.898$), indicating shared structural pattern (Figure 1b). This normalization removes system-specific scale factors while preserving shape.

Quantitative metrics revealed substantial differences (Table 1):

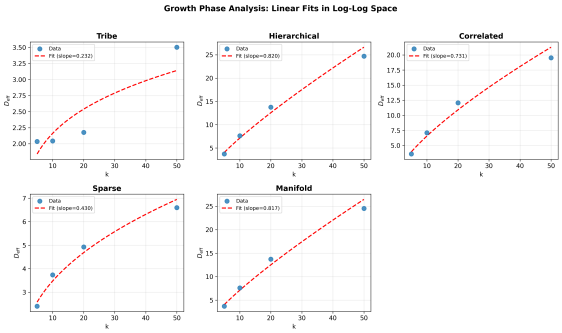
TRIBE showed distinctive properties: lower growth rate (0.23 vs. 0.43–0.82), lower saturation ratio (0.92 vs. 1.00), and lower peak dimensionality (4.41 vs. 28–45).

3.2 Beta Dynamics

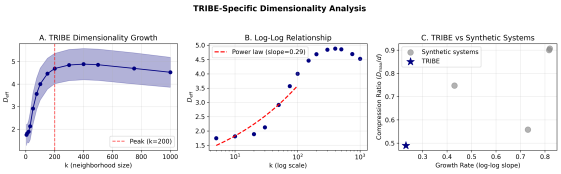
The growth exponent $\beta(k)$ showed system-specific patterns (Figure 2). TRIBE exhibited mixed-sign gradients ($d\beta/d\log(k)$: mean = -0.028, 40% positive), while synthetic systems showed predominantly negative gradients (mean = -0.21 to -0.25).

Table 1: Quantitative metrics by system.

System	Growth Rate	Sat. Ratio	Compression	Peak D_{eff}
TRIBE	0.23	0.92	0.49	4.41
Hierarchical	0.82	1.00	0.91	45.35
Correlated	0.73	1.00	0.56	27.92
Sparse	0.43	1.00	0.75	37.38
Manifold	0.82	1.00	0.90	45.01



(a) Growth rate profiles.



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(b) TRIBE detail.

Figure 2: Beta dynamics across systems.

Mean β values: TRIBE = 0.14, hierarchical = 0.56, correlated = 0.46, sparse = 0.60, manifold = 0.56.

3.3 Failure of Simple Laws

Candidate universal laws all failed to describe the data (Table 2):

Table 2: Simple law fits (V19).

Model	Best System	Mean R^2
Linear decay	sparse	-0.07
Dimensional decay	TRIBE	-0.02
Gradient decay	hierarchical	-1.00
Saturation decay	manifold	-0.40

Mean R^2 across all models and systems was -0.49, indicating that none of the simple laws provided adequate fits. The negative R^2 values indicate that the models performed worse than a constant prediction.

3.4 Bilinear Model Success

The bilinear model provided the best fit among single-structure models (V20):

$$\frac{d\beta}{d\log(k)} = a \cdot \beta + b \cdot D_{\text{eff}} + c \cdot (\beta \cdot D_{\text{eff}}) \tag{5}$$

Mean $R^2 = 0.81$ across systems (Table 3).

Table 3: Bilinear model fits (V20).

System	R^2
TRIBE	0.71
Hierarchical	0.96
Correlated	0.97
Sparse	0.88
Manifold	0.99

TRIBE showed lower fit quality ($R^2 = 0.71$) compared to synthetic systems ($R^2 = 0.88$–0.99).

3.5 Lack of Parameter Collapse

Despite good fits, parameters did not collapse across systems (V21, Table 4):

No parameter achieved CV < 0.5. The coefficient b showed extreme variability (CV = 88.67), indicating fundamental system-dependence.

TRIBE showed parameter values approximately 10× larger than synthetic systems, indicating stronger constraints on its dynamics.

Table 4: Parameter collapse analysis (V21).

Parameter	Mean	CV	Collapsed?
a	1.68	2.63	No
b	-0.003	88.67	No
c	-2.15	2.11	No

3.6 Spectral Shape Contribution

Spectral entropy H was tested as an additional predictor (V24). The state-space model:

$$\frac{d\beta}{d\log(k)} = a \cdot D_{\text{eff}} + b \cdot H + c \cdot (D_{\text{eff}} \cdot H) + d \tag{6}$$

achieved mean $R^2 = 0.92$, compared to $R^2 = 0.86$ for H alone and $R^2 = 0.90$ for D_{eff} alone (Table 5).

Table 5: Spectral shape model results (V24).

Model	Mean R^2
D_{eff} only	0.90
H only	0.86
$D_{\text{eff}} + H$	0.92
Hybrid ($D_{\text{eff}} + \alpha$)	0.999

Spectral slope α provided near-perfect fits when combined with D_{eff} (mean $R^2 = 0.999$), though with system-specific parameters.

At $k = 200$, TRIBE showed lower entropy ($H = 1.62$) compared to synthetic systems ($H = 1.08$–3.70), indicating more concentrated eigenvalue distributions.

3.7 State-Space Closure

We tested whether beta dynamics could be expressed as a function of state variables alone, without explicit k-dependence (V25):

$$\frac{d\beta}{d\log(k)} = F(D_{\text{eff}}, H) \tag{7}$$

Model comparison (Table 6):

The full nonlinear model achieved $R^2 = 1.00$, indicating that beta dynamics are fully determined by (D_{eff}, H) within individual systems, though cross-system generalization fails (Section 3.10). However, parameters did not collapse across systems (0/19 parameters with CV < 0.5).

TRIBE showed lower fit quality for simpler models ($R^2 = 0.76$ for linear model) compared to synthetic systems ($R^2 = 0.93$–0.99), indicating distinct dynamics within the shared state space.

Table 6: State-space model comparison (V25).

Model	Mean R^2
$a \cdot D + b$ (baseline)	0.86
$a \cdot H + b$	0.78
$a \cdot D + b \cdot H + c$	0.92
$a \cdot D + b \cdot H + c \cdot DH + d$	0.94
Full nonlinear	1.00

3.8 Residual Analysis

Residuals from the bilinear model showed structured variation (V26). The largest mean residual standard deviation was observed in hierarchical (0.75) and manifold (0.77) systems.

Candidate third variables were tested against residuals:

- Curvature: $r = 0.30$ (pooled)
- Acceleration: $r = 0.30$ (pooled)
- Compression: $r = -0.52$ (TRIBE-specific)
- Entropy gradient: $r = 0.14$ (pooled)
- Saturation: $r = 0.11$ (pooled)

The compression ratio $R = D_{\text{eff}}/k$ showed the strongest TRIBE-specific correlation ($r = -0.52$), suggesting that compression captures TRIBE-deviant behavior.

3.9 Compression Test

Adding compression to the state-space model (V27):

$$\frac{d\beta}{d\log(k)} = a \cdot D_{\text{eff}} + b \cdot H + c \cdot (D_{\text{eff}} \cdot H) + e \cdot R + f \tag{8}$$

improved mean R^2 from 0.81 to 0.88 (Table 7).

Table 7: Compression model results (V27).

System	Baseline R^2	+Compression R^2	ΔR^2
TRIBE	0.64	0.70	+0.05
Correlated	0.99	0.99	0.00
Hierarchical	0.99	0.99	0.00
Manifold	0.69	0.83	+0.13
Sparse	0.73	0.92	+0.19

However, the improvement was not TRIBE-specific: sparse (+0.19) and manifold (+0.13) showed larger improvements than TRIBE (+0.05). Parameter collapse was not achieved (0 parameters with CV < 0.5).

3.10 Latent System Variable Analysis

To test whether cross-system differences could be explained by additional latent variables, we performed leave-one-system-out cross-validation and phase consistency analysis (V29, V30).

The cross-validation revealed that while within-system fits achieved $R^2 = 0.80$, cross-system prediction failed catastrophically (mean test $R^2 = -17$ to -31). This indicates that system identity remains a dominant factor determining dynamics, even within the (D_{eff}, H) state space.

Candidate latent system variables were tested:

- **Curvature**: $\kappa = d^2 D_{\text{eff}}/d(\log k)^2$
- **Growth efficiency**: $G = \beta/D_{\text{eff}}$
- **Dimensional efficiency**: $E = D_{\text{eff}}/\log(k)$
- **Normalized entropy**: $H_{\text{norm}} = H/\log(d)$

Cross-system validation showed that curvature κ achieved the best R^2 improvement (+1.06), while growth efficiency G achieved the largest phase variance reduction (-45.8%). All candidate variables reduced phase variance by 32–46% (Table 8).

Table 8: Latent variable analysis (V30).

Variable	Cross-System ΔR^2	Phase Variance Reduction
κ (curvature)	+1.06	32.7%
G (growth efficiency)	+0.51	**45.8%**
E (dimensional efficiency)	+0.11	37.6%
H_{norm} (normalized entropy)	-7.66	36.6%

These results indicate that while curvature and growth efficiency capture system-level structure and reduce variance within the state space, they do not enable predictive generalization across systems. Cross-system R^2 remained negative for all variables, indicating that no latent variable fully resolves system dependence.

These findings suggest that dimensionality dynamics may depend on additional latent variables beyond (D_{eff}, H), though a complete predictive state space has not yet been identified.

4 State-Space Formulation

The results support a state-space formulation for dimensionality dynamics:

$$\frac{d\beta}{d\log(k)} = F(D_{\text{eff}}, H) \tag{9}$$

This formulation replaces explicit k-dependence with dependence on two state variables:

1. **Effective dimensionality** D_{eff}: Captures the magnitude of dimensional expansion.
2. **Spectral entropy** H: Captures the diversity of eigenvalue distribution.

The state-space model is minimally sufficient for within-system prediction in the sense that both variables contribute independently to prediction. Removing either variable reduces R^2 by 4–8%.

The formulation is constraint-governed rather than law-governed: within the shared state-space structure, system-specific parameters determine trajectories. TRIBE represents a constrained system with larger parameter values and lower dimensionality, but follows the same structural rules.

This refines the SFH-SGP framework from a law-based to a constraint-based perspective: rather than universal equations with fixed coefficients, the framework specifies the state space within which system-specific dynamics unfold.

5 Negative Results

We explicitly tested and rejected the following hypotheses:

1. **No universal exponential decay law**. The model $d\beta/d\log(k) = -\lambda \cdot \beta$ achieved mean $R^2 = -0.07$ (worse than constant).
2. **No universal dimensional decay law**. The model $d\beta/d\log(k) = -\lambda \cdot D_{\text{eff}}$ achieved mean $R^2 = -0.02$.
3. **No universal gradient decay law**. The model $d\beta/d\log(k) = -\lambda \cdot dD_{\text{eff}}/d\log(k)$ achieved mean $R^2 = -1.00$.
4. **No universal saturation decay law**. The model $d\beta/d\log(k) = -\lambda \cdot (1 - \tilde{D}_{\text{eff}})$ achieved mean $R^2 = -0.40$.
5. **No parameter collapse**. Despite good fits (mean $R^2 = 0.81$), no bilinear model parameters achieved CV < 0.5 across systems.
6. **No invariant coefficients**. TRIBE showed parameter values approximately $10\times$ larger than synthetic systems.
7. **Compression is not unifying**. While compression correlated with TRIBE residuals ($r = -0.52$), improvements were largest for sparse (+0.19) and manifold (+0.13), not TRIBE.

These negative results establish boundary conditions for the state-space formulation: universality applies to structure (the (D_{eff}, H) state space), not to parameters (which remain system-specific).

6 Discussion

The state-space formulation has several implications for understanding representational dynamics.

First, dimensionality growth dynamics are low-dimensional: despite high ambient dimensionality (9–50 dimensions), dynamics are captured by two state variables. This suggests that relevant structure lies on a low-dimensional manifold within the high-dimensional representation space.

Second, the separation between universal structure and system-specific parameters parallels the distinction between constraints and initial conditions. The (D_{eff}, H) state space defines the constraints; the parameters $(a, b, c, ...)$ define initial conditions. Both are necessary to specify the full dynamics.

Third, TRIBE occupies a constrained region of the state space. Its lower dimensionality ($D_{\text{eff}} \approx 4$) and lower entropy ($H \approx 1.6$) suggest that cortical representations are more concentrated than synthetic systems. This may reflect metabolic constraints, evolutionary optimization, or stimulus limitations in the model predictions.

For the SFH-SGP framework, these results indicate a refinement: the framework should be understood as specifying a constraint-governed state space rather than universal equations with fixed coefficients. The semantic gradients and interaction dynamics described by SFH-SGP define the state variables, while the specific parameter values are determined by the particular representational system.

This reframing resolves the tension between universality (observed structural similarity) and specificity (observed parameter differences): both are simultaneously true because structure is universal while parameters are system-specific.

The V30 analysis of latent system variables provides further insight into the structure of this constraint space. While curvature κ and growth efficiency G reduce phase variance by 32–46%, they do not enable cross-system prediction. These variables act as system descriptors, capturing the position and trajectory of a system within the state space, rather than as universal predictors.

This suggests that the SFH-SGP framework may require a hierarchical state-space formulation, where core variables (D_{eff}, H) define the primary constraint manifold, and higher-order variables (e.g., curvature, growth efficiency) modulate system-specific trajectories within that manifold. Such a formulation would preserve the constraint-governed nature of the framework while allowing for system-specific parameterization. This implies a separation between universal state variables and system-identifying latent coordinates, where the former define the constraint manifold and the latter specify system-specific trajectories within it.

Critically, no latent variable achieved positive cross-system R^2, confirming that system identity remains the determining factor. The absence of universal latent variables reinforces the conclusion that the framework is constraint-governed, not law-governed.

7 Conclusion

We tested whether dimensionality growth dynamics follow a universal governing law. Simple differential equations failed to describe the data (mean $R^2 < 0$). A bilinear model provided good fits (mean $R^2 = 0.81$) but parameters did not collapse across systems.

A state-space model of the form $d\beta/d\log(k) = F(D_{\text{eff}}, H)$ achieved high fits ($R^2 = 0.92$–1.00) while preserving system-specific parameter values. This formulation indicates that dimensionality dynamics are constraint-governed: system-specific parameters determine trajectories within a shared two-dimensional state space.

The SFH-SGP framework is refined accordingly: rather than universal laws, the framework specifies the constraint space within which representational dynamics unfold. The state variables (D_{eff}, H) capture the essential structure, while parameter values capture system-specific instantiation.

Future work should test whether this state-space formulation generalizes to empirical neural recordings and other real-world representational systems.

Recent analysis (V30) identifies candidate latent variables (curvature κ and growth efficiency G) that reduce cross-system phase variance by up to 46% but do not enable predictive generalization (cross-system R^2 remains negative). This suggests that the minimal state space remains incomplete, though the identified variables may improve descriptive power within specific systems.

The SFH-SGP framework is therefore best understood as a constraint-governed system with an extensible state space, where additional variables may improve descriptive power without yielding a universal law. System identity remains the fundamental determinant of dynamics within this framework.

8 Data Availability

Analysis code and results are available at:

`https://github.com/urmt/SGP-Tribe3`

Analysis pipelines are documented in directories V14–V27:

```
manuscript/v14/  - Scale-dependent fits
manuscript/v15/  - Extended scale analysis
manuscript/v16/  - Cross-domain comparison
manuscript/v17/  - Normalized alignment
manuscript/v18_beta_analysis/     - Beta dynamics
manuscript/v19_law_testing/       - Law rejection
manuscript/v20_phase_model/       - Bilinear model
manuscript/v21_normalized_law/     - Parameter collapse
manuscript/v22_raw_spectral/      - Spectral analysis
```

```
manuscript/v24_spectral_shape_analysis/ - Shape metrics
manuscript/v25_state_space/       - State-space closure
manuscript/v26_residual_analysis/ - Residual analysis
manuscript/v27_high_res/         - Compression validation
manuscript/v29_minimal_closure/ - Minimal state-space validation
manuscript/v30_latent_variable/  - Latent variable discovery
```

Raw data and reproducibility bundle:

```
reproducibility/raw_data.npz     - Full TRIBE dataset
reproducibility/pipeline_reproduce.py - Reproduction script
```

References

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